

Lecture 7 Team Projects

Interest-Rate Models

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1 Project 1: Vasicek as a Risk Scenario Generator

Scenario. Explain what a mean-reverting short-rate model can and cannot say about future rates.

Objectives. Choose defensible parameters; interpret mean reversion and volatility; compare one-year model distributions with historical changes.

Sources. FRED or Bank of Canada short rates; instructor monthly series and simulation template.

Deliverables. Parameter rationale, distribution and three paths, historical comparison, model-risk memo, and a 10–12 minute presentation.

Rubric (100 points). Parameters 25; interpretation 25; scenarios 20; validation and model risk 20; presentation 10.

2 Project 2: Why Hull–White Must Fit Today’s Curve

Scenario. Evaluate a proposal to price a callable bond without first matching current market prices.

Objectives. Distinguish equilibrium and no-arbitrage models; demonstrate the importance of curve fit; interpret mean-reversion and volatility effects.

Sources. U.S. Treasury, FRED, or Bank of Canada curve data; instructor Hull–White tree or calibration output.

Deliverables. Model comparison, curve-fit exhibit, two sensitivities, instrument-specific recommendation, and a 10–12 minute presentation.

Rubric (100 points). Model distinction 25; curve-fit reasoning 25; sensitivities 20; recommendation 20; presentation 10.

3 Project 3: Model Choice for an ALM Committee

Scenario. Choose between historical scenarios, Vasicek, and Hull–White for a pension plan’s asset-liability analysis.

Objectives. Match each method to purpose; compare calibration, curve consistency, interpretability, and tail behavior; recommend a primary model and challenger.

Sources. Public pension disclosures; FRED or Bank of Canada curves; instructor scenario outputs.

Deliverables. Requirements matrix, three-model comparison, scenario evidence, governance recommendation, and a 10–12 minute presentation.

Rubric (100 points). Problem framing 20; model comparison 30; evidence 20; governance and recommendation 20; presentation and sourcing 10.

4 Project 4: Stress-Test Mean Reversion

Scenario. A risk manager doubts that a single estimated mean-reversion speed is stable enough for decisions.

Objectives. Estimate parameters over contrasting periods; interpret estimation uncertainty; test portfolio values under slow, base, and fast mean reversion.

Sources. FRED or Bank of Canada monthly short rates; instructor bond cash flows and spreadsheet model.

Deliverables. Sample-period comparison, parameter table, three valuation scenarios, limitation note, risk-limit recommendation, and a 10–12 minute presentation.

Rubric (100 points). Data design 20; parameter reasoning 25; stress analysis 25; decision implications 20; presentation and reproducibility 10.